

SYNOPSIS

At 1605 on 15 April 2015, a dived Royal Navy submarine snagged the fishing gear of the UK registered trawler *Karen*, 15 miles south-east of Ardglass, Northern Ireland. *Karen* had been trawling for prawns on a westerly heading at 2.8 knots when its fishing gear was snagged and it was dragged backwards at about 7 knots. *Karen*'s crew managed to release both winch brakes, freeing the trawl warps; the starboard warp ran out completely but the port warp became fouled on the winch drum, causing *Karen* to heel heavily to port and its stern to be pulled underwater. *Karen* broke free from the submarine when the port warp parted; there was structural damage to the vessel but it returned to Ardglass safely under its own power. Evidence of the collision on board the submarine was either not observed or misinterpreted. This meant that the submarine did not render immediate assistance as the command team was unaware of the collision until about 3 hours later.

The collision occurred because the submarine's command team assessed that *Karen* was a merchant vessel, primarily because no trawl noise¹ was heard. The submarine was at a depth where it could, if necessary, pass safely beneath a merchant vessel, therefore the command team would not have perceived any risk of collision; as a result, no avoiding action was taken.

The submarine's command team had assessed that the majority of shipping contacts in the area were merchant vessels. However, most were actually trawlers; this was predictable and should have been identified as a significant risk to the safety of the submarine and other vessels when preparing the submarine's passage plan. Had the submarine's command team appreciated the high density of fishing vessels and then followed Royal Navy guidance on fishing vessel avoidance, the accident would have been avoided because the submarine would have been slowed down and returned to periscope depth when the density of shipping increased.

This investigation was conducted without the full co-operation of the Royal Navy. The involvement of a submarine was not revealed until nearly 5 months after the accident and it took 10 months for the Royal Navy to submit evidence to the investigation team. These delays impeded the progress of the independent investigation, and the evidence submitted was insufficient to determine all the causal factors.

This report makes safety recommendations to the Royal Navy to review the procedures and training necessary to ensure that submarine operations in the vicinity of vessels engaged in fishing are conducted safely, and to provide assurance that actions have been taken to prevent recurrence.

¹ See Section 1.8.1.

SECTION 1 – FACTUAL INFORMATION

1.1 PARTICULARS OF *KAREN*, THE SUBMARINE AND THE ACCIDENT

SHIP PARTICULARS		
Vessel's name	<i>Karen</i>	Not declared
Flag	United Kingdom	United Kingdom
IMO / Fishing number	B317	Not declared
Type	Stern trawler	Submarine
Registered owner	T Wills and Son (Northern Ireland) Ltd	UK Ministry of Defence
Construction	Wood	Steel
Year of build	1975	Not declared
Length overall	19.23m	Not declared
Registered Length	17.71m	Not declared
Gross tonnage	50	Not declared
Authorised cargo	Fish	Not applicable
Port of departure	Ardglass, Northern Ireland	HMNB Clyde, Faslane
Intended port of arrival	Ardglass, Northern Ireland	HMNB Clyde, Faslane
Type of voyage	Commercial fishing	Military operations
Cargo information	Prawns	Not applicable
Manning	4	Not declared
MARINE CASUALTY INFORMATION		
Date and time	15 April 2015, 1605	
Type of marine casualty or incident	Serious Marine Casualty	
Location of incident	54°03.3'N - 005°20.4'W	
Place on board	Ship	Not declared
Injuries/fatalities	None	None reported
Damage/environmental impact	Net, trawl warps and trawl doors lost. Structural damage to port gallows, deck and hydraulic winch bedplate crossbeam.	None reported
Ship operation	Engaged in fishing	Military operations
Voyage segment	Mid-water	Mid-water
External & internal environment	Wind: westerly, force 3 Visibility: good Sea State: slight	
Persons on board	4	Not declared

1.2 NARRATIVE

1.2.1 Events prior to the collision

In early April 2015, a pre-deployment brief was held on board a Royal Navy submarine (the submarine) while it was berthed at Her Majesty's Naval Base, Clyde in Faslane, Scotland (HMNB Clyde). At the briefing, which was attended by the captain of the submarine flotilla based in Faslane, consideration was given to potential routing options for the submarine's departure from UK waters. During this discussion, the submarine's Commanding Officer stated his intention to conduct a dived transit south through the Irish Sea.

At an undisclosed time, the submarine sailed from HMNB Clyde, dived, then conducted routine equipment performance checks. Once these checks were complete, the submarine commenced its dived passage south towards the Irish Sea, proceeding below periscope depth² (**Figure 1**).

At 0430 on 15 April 2015, the stern trawler *Karen* sailed from Ardglass, Northern Ireland and headed out to sea. When *Karen* reached its intended fishing grounds, the crew shot the nets and, about 3 hours later, hauled the gear and landed the first catch. This process was repeated and, at about 1500, the crew landed their second haul of the day and shot the net for the third time. Once the gear was away, the skipper settled the vessel on an autopilot controlled heading of 267° at a speed over the ground of about 2.8 knots (kts). The skipper then kept watch in the wheelhouse while the rest of the crew processed the previous catch.

During the afternoon, as the submarine progressed its passage south, the Commanding Officer noted that the traffic density had increased; most of the sonar contacts were assessed to be merchant vessels. The submarine's Commanding Officer considered that it would be impractical to conduct a close quarters procedure³ for every contact so a decision was taken to continue the dived transit, accepting that the submarine would pass close to merchant vessels.

At 1556, the submarine was on a south-westerly heading at 10kts and the sonar operators identified a new contact, fine on the port bow, that had previously been masked by several other vessels (**Figure 2**); this was *Karen*. In order to determine the contact's range, the submarine's heading was altered 15° to starboard⁴. The sonar operators listened to *Karen* and analysed the acoustic information; this included an assessment of its shaft revolutions per minute (rpm), propeller configuration and aural characteristics. Based on this information and the absence of trawl noise, the submarine's command team assessed that *Karen* was a small merchant vessel. The submarine's heading was then altered to port onto a south-easterly course in order to determine the ranges of other passive sonar contacts.

² See Section 1.7.3 for description of periscope depth.

³ A close quarters procedure is a rapid, predetermined set of reactions taken by a submarine to avoid a collision. See also Section 1.8.3.

⁴ A description of how a submarine uses passive sonar to detect, locate and classify vessels is at Section 1.8.

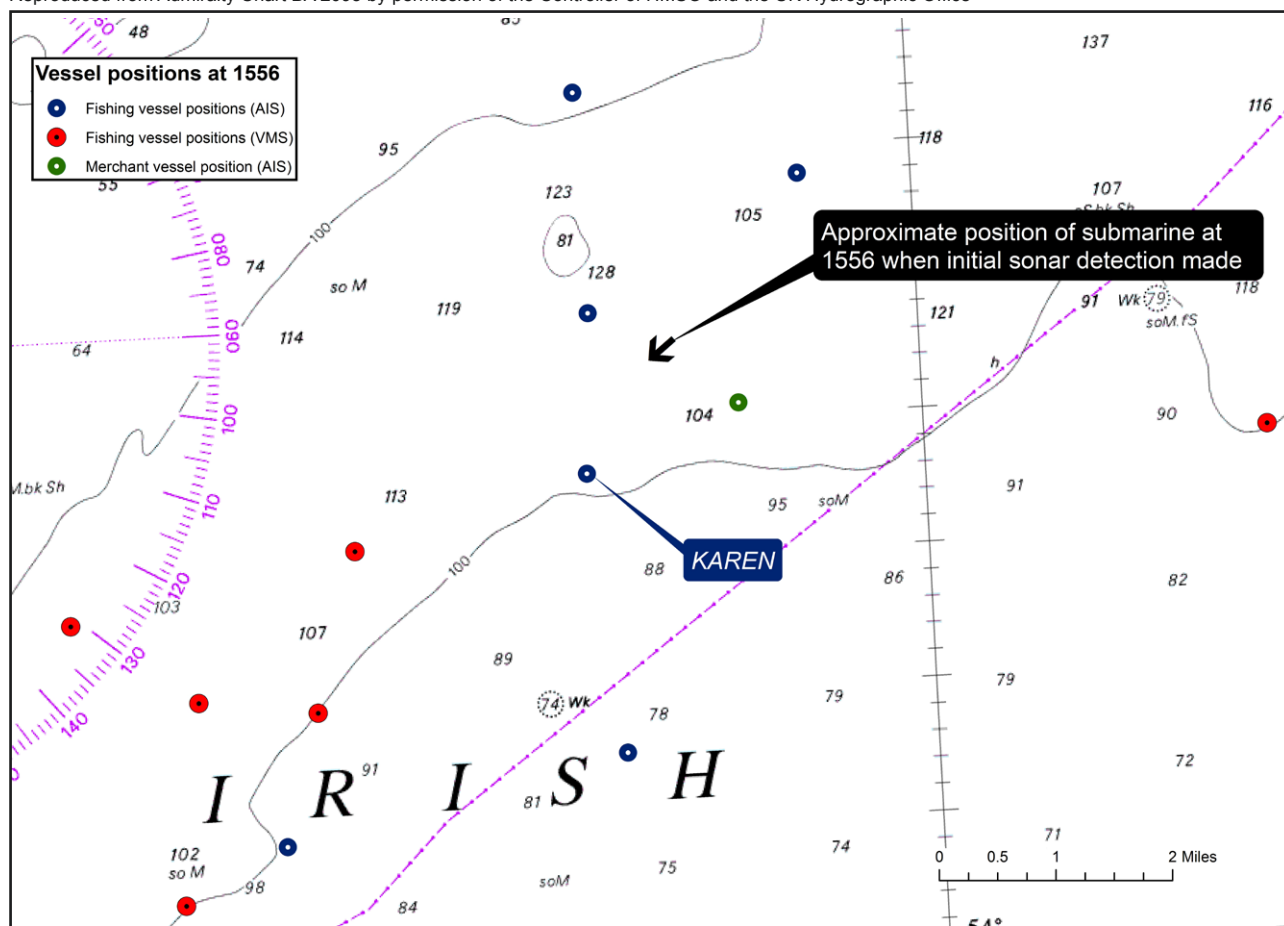


Figure 2: Plot showing the approximate position of the submarine and adjacent shipping at 1556, based on AIS and VMS data

1.2.2 The collision

At 1605, having steadied on a south-easterly heading, the submarine passed close to *Karen's* stern and snagged its fishing gear. When the collision occurred, *Karen's* crew heard an unusual noise and realised something was wrong as the trawl warps had unexpectedly tightened and the vessel started to be dragged backwards (**Figure 3**). The skipper immediately disengaged the propeller and shouted to the crew to release the winch brakes; having done that, both warps started to run away freely. The starboard warp ran out completely but the port warp became fouled on its winch drum and again came taut. This caused *Karen* to heel to port and ship water over the stern as the vessel started submerging backwards under the downward pull on the port warp. Unable to release the tension on the port warp, one of the crewmen started preparing the liferaft for launch, while the others kept clear of the taut wire. After being dragged backwards at about 7kts for around 30 seconds, the port warp parted and *Karen* was released.

1.2.3 Post-collision

After the port warp parted, *Karen* slowed down, returned to upright and the water drained from the deck. With the fishing gear gone, the crew began to assess the situation on board; damage was evident to the port gallows, but the engine was still running and there was no evidence of internal flooding.

At about 1610, the chief petty officer in charge of the sonar team on board the submarine went to the control room to brief the command team that an unusual noise had been detected. Although only audible for a short duration, the noise was

about submarine operations. Commanding Officers of submarines also held orders in operational documentation that normally required them to conduct operations without being detected; one aspect of this would be to maintain radio silence¹¹.

1.7.3 Modes of operation

Submarines can operate in two modes:

- Surfaced: The submarine is fully buoyant with the fin and casing visible and an officer of the watch / lookout posted on the bridge.
- Dived:
 - Periscope depth: The submarine is fully submerged but at a shallow depth where periscopes and radar could be used.
 - Deep: Any depth below periscope depth.

When surfaced, submarines comply with the International Regulations for the Prevention of Collisions at Sea (COLREGs), including showing appropriate lights and shapes. There is no provision in the COLREGs for dived submarines; this means that responsibility for collision avoidance rests entirely with its commanding officer once a submarine has dived. When deep, submarines operate at a depth where there is no risk of collision with any merchant vessel that could be encountered; this is known as 'safe depth' (**Figure 14**).

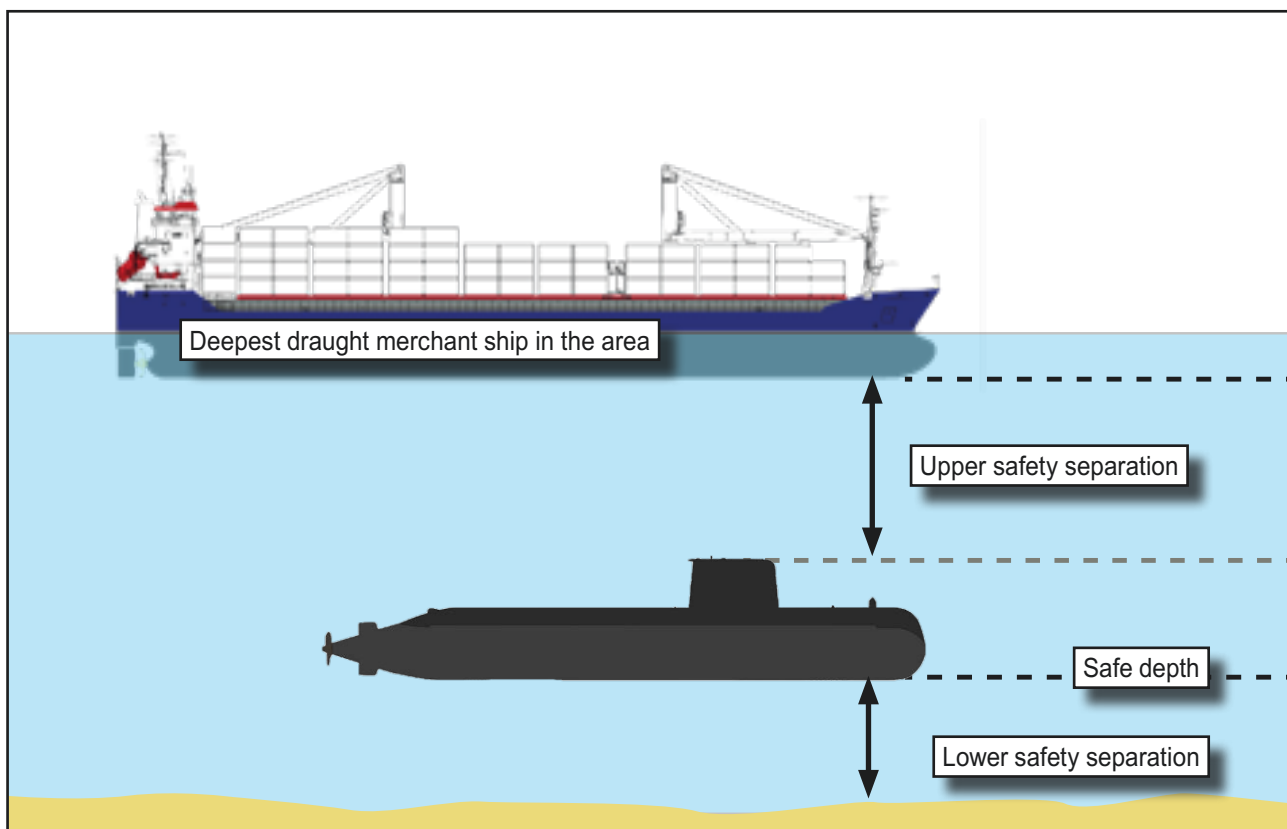


Figure 14: Illustration (not to scale) of a submarine at safe depth

¹¹ 'Radio silence' meaning receiving communications from shore but not transmitting any messages back by any means.

The bearings of passive contacts are plotted by a computer and this data can be used by tactical system operators to estimate the range, course and speed of the ship being detected. This process is called target motion analysis (TMA) and an estimated position, course and speed of the other vessel is referred to as a *solution*.

The process of determining the range of a passive sonar contact can be significantly improved by the submarine altering course and/or speed as the subsequent change in bearing movement allows the command team to refine the solution, in particular the contact's range. Changing the submarine's course and/or speed to determine the range of passive contacts is called a *ranging manoeuvre*.

The TMA process can also be refined using the additional acoustic evidence available. For instance, if a passive sonar contact has the acoustic characteristic of a large merchant vessel, the submarine's command team can work to find solutions at speeds typical for such a vessel. Equally, if the sonar operators hear trawl noise, then the command team will classify the contact as a vessel engaged in fishing, and find TMA solutions at speeds typical for such a vessel. The workload of sonar and tactical system operators is a function of the number of contacts being detected and analysed.

Figure 15 illustrates the passive TMA process prior to the collision. The submarine's track, including the ranging manoeuvre, is shown in black and two possible TMA solutions are also shown. The more distant contact (shown in green) is a solution using a typical merchant vessel speed of 10kts, and the closer contact (shown in red) is a solution for a vessel speed of 2.8kts, representative of *Karen's* actual relative movement.

1.8.3 Close quarters procedures

In the event of a deep submarine making a sonar detection of a new contact that could present a risk to the submarine, the command team would undertake a *close quarters procedure*. This was a rapid, predetermined set of reactions intended to avoid collision. The Royal Navy has not disclosed to the MAIB the nature of the close quarters procedures in use by its submarines or the detail of the circumstances in which such action would be taken.

1.9 PASSAGE PLANNING

1.9.1 Royal Navy guidance

The Admiralty Manual of Navigation Volume 1, *The Principles of Navigation*, Book of Reference (BR) 45(1) is the Royal Navy's primary reference for the planning and conduct of navigation. The conduct of coastal navigation is covered by Chapter 12; key extracts include:

SECTION 1 – PLANNING COASTAL NAVIGATION

1210. Check Lists and Navplan Preparations

b. Appraisal. After completing initial research, appraisal of the following items should be carried out as soon as possible thereafter...

- *ETD¹²s / ETA¹³s and refined speed of advance (SOA).*
- *TSS¹⁴, shipping lanes, traffic density, likely concentrations of fishing vessels.*
- *Exercise areas.*
- *Any limitations of ship.*
- *Maritime Jurisdiction, Innocent Passage and Diplomatic Clearance.*
- *Intelligence requirements / planning.*
- *Cross check all available sources of navigation and other information.*

1211. Criteria for Route Selection

Fishing Vessels. *In Coastal Navigation, fishing vessels / fishing fleets may be encountered. Concentrations of fishing vessels should be avoided if possible.*

Additional guidance was available to submarine navigating officers for planning dived submarine navigation; this information was not released to the MAIB.

1.9.2 Fishing patterns

On the day of the accident, *Karen* was one of 61 fishing vessels operating in the fishing grounds between Ireland and the Isle of Man. At the time of the accident, this area was included in the European Union (EU) *Cod Recovery Zone*¹⁵, active between 14 February and 30 April 2015. This regulation prevented any type of fishing for white fish and, as a result, all of the active trawlers in the area would have been fishing the seabed for prawns. Prawns can be caught all year round, but spring and summer provide a better harvest due to the longer daylight hours. This was relevant for vessels such as *Karen* that use a net, designed to catch prawns that have emerged from seabed burrows in daylight. Larger fishing vessels with heavier gear can work all hours as the trawls dig into the seabed and catch prawns in any light condition. This meant that the concentration of vessels engaged in fishing in the area was likely to reduce during darkness. In addition, between October and May, the prawn fishery is generally better in deeper water. Typically, this results in a situation where submarines and trawlers will both tend to operate in the same area during the winter and spring months.

When active at fishing grounds, trawlers are normally in a continuous process of shooting nets, trawling and then recovering the gear, thus the risk of collision for a submarine operating in the same area is ever-present.

¹² Estimated time of departure.

¹³ Estimated time of arrival.

¹⁴ Traffic separation schemes.

¹⁵ The Cod Recovery Zone in the Irish Sea was implemented by the UK Government in order to meet the requirements of the EU Council Regulations No. 1342/2008 and 57/2011.

1.10 SUBMARINE OPERATIONS IN THE VICINITY OF FISHING VESSELS

1.10.1 Guidance for submarine command teams

Guidance for UK submarine command teams was contained in the Royal Navy's BR0095, *Fishing Vessel Avoidance*; key extracts include:

Annex 1A stated that *the safety of life is of paramount importance and over-rides all other considerations.*

Para 0103(b) stated that *where practicable routine transits of coastal waters will be undertaken on the surface, but that essential dived transits through coastal waters are to be conducted at periscope depth (PD) at slow transit speeds and only for short periods...*

Para 0104(5) specified a mandatory separation of at least 4000yds¹⁶ from all vessels classified as possible fishing vessels, whether or not they are believed to be engaged in fishing.

Para 0105(a) stated that *when shipping and navigation constraints arise, and it becomes impractical, for whatever reason, for a submarine below PD to maintain 4000 yards separation, it must return to PD in as short a time as possible.*

1.10.2 The Fishing Vessel Code of Practice

In 1993, The Royal Navy first published its unclassified *Code of Practice for Fishing Vessel Avoidance* (the FV Code) (**Annex A**).

The FV Code had been accepted by the Fishing Industry Safety Group (FISG) and was last revised and updated in 2002. It stated that the FISG had acknowledged an *essential military requirement* to operate submarines dived in waters where UK vessels engaged in fishing may be operating. Similar to BR0095, the FV Code stated that, if a deep submarine was unable to maintain a mandatory separation of 4000yds from any vessel possibly engaged in fishing, it should return to periscope depth as soon as possible.

In the event of a collision between a submarine and a fishing vessel, the FV Code listed actions to be taken by the submarine; these included slowing down, returning to periscope depth, communicating with the fishing vessel, surfacing to render assistance and reporting the incident to its Operating Authority.

1.10.3 Submarine information broadcasts

The Royal Navy operated a warning system, referred to as SUBFACTS¹⁷, intended to notify mariners of the presence of dived submarines operating in designated exercise areas (**Figure 16**). SUBFACTS information was compiled by the duty submarine controller at the Fleet Operations Division in Northwood. It was then broadcast at 0620 UTC and 1820 UTC daily via the coastguard on VHF radio and also using the Navigational and Meteorological Warning Broadcast Service (NAVTEX).

¹⁶ Distances in this report relating to submarine operations are stated in yards, which is the unit of measurement used by UK submarines; 1 yard equals 0.9144m.

¹⁷ Full details of the SUBFACTS system were promulgated in the Admiralty List of Radio Signals (ALRS) Volume 3 Part 1.

The accident location was about 1 mile inside the southern boundary of this SUBFACTS promulgated area.

Similar messages were transmitted the day before the accident and later the same day, which warned of submarine activity in the same areas between 0700 on 14 April 2015 until 0800 on 16 April 2015.

1.10.4 Marine Guidance Note 12(F)

In 1997 the Marine Safety Agency (MSA)¹⁹ published its Marine Guidance Note (MGN) 12(F), *Fishing vessels operating in submarine exercise areas (Annex B)*. The MGN described the measures implemented by the Royal Navy to minimise the risks to fishing vessels when operating in known submarine exercise areas. This included details of the exercise areas, the SUBFACTS broadcast arrangements and the responsibilities of the Fishing Vessel Safety Ship (FVSS) appointed during submarine exercises. MGN 12(F) also contained recommended actions to be taken by fishing vessels when operating within a known submarine exercise area. This advice included:

- Making use of the SUBFACTS broadcast.
- Ensuring a radio watch is kept on VHF channel 16.
- If stopped in the water, ensuring all electronic equipment such as fish finders and echo sounders, capable of generating pulses in the water, are operating.
- In the event of a total power failure and, therefore, a risk of not being detected by a submarine's passive sonar, the fishing vessel should contact the coastguard immediately and consider releasing fishing gear for later recovery.

The MGN also reminded fishing vessel skippers to ensure that they are showing the correct navigation lights and shapes in order to allow dived submarines at periscope depth to correctly identify them.

1.10.5 Military exercises

The military exercise taking place at the time of the accident was Exercise *Joint Warrior 151*. The MoD's Joint Tactical Exercise Planning Staff (JTEPS) plan and deliver this major land, sea and air training exercise that takes place twice a year, predominantly in areas to the north and west of Scotland. Before the exercise started, the Director of JTEPS published a letter (**Annex C**) setting out the exercise areas in use, including where dived submarine operations were planned to take place. The submarine exercise areas detailed in the JTEPS letter (**Annex C**) did not correlate with the areas published in MGN 12(F) and the areas it indicated as being active on 15 April 2015 were not included in the SUBFACTS broadcast for that day.

¹⁹ MGN12(F) was published in 1997 by the Marine Safety Agency and, although this predates the formation of the Maritime and Coastguard Agency, this MGN remains in force as an MCA publication.

- As the submarine was deep and could not make a visual classification, *Karen's* contact should have been treated as a fishing vessel.
- The submarine crew were operating near to their limit of capability.
- The Commanding Officer could have taken action to reduce workload by reducing the submarine's speed of approach and returning to periscope depth.
- The Operating Authority's post-incident signal, informing the submarine of the reported incident, should have required a positive response had the submarine been involved.
- There was no evidence of any equipment defect that would have prevented safe dived operations during the submarine's transit through the Irish Sea.

The Directed Review report listed the actions already taken by the Royal Navy and included a number of recommendations aimed at reducing the likelihood and potential consequences of collisions with fishing vessels in the future.

1.14 PREVIOUS OR SIMILAR ACCIDENTS

1.14.1 Collision between HMS *Trenchant* and *Antares*

At 0217 on 22 November 1990, the Royal Navy Trafalgar Class nuclear powered submarine, HMS *Trenchant*, collided with the trawl gear of the pelagic trawler *Antares* east of the Island of Arran, Scotland. The collision resulted in *Antares* capsizing and foundering with the loss of all four crew. The MAIB investigation²⁵ established that the collision was caused by a breakdown in the watchkeeping structure and standards on board the submarine. The submarine's command team had no clear appreciation of the surface contacts held on sonar prior to the collision, and the absence of trawl noise from *Antares* led to an incorrect assumption that the contact was not engaged in fishing.

After the collision, a sonar operator on board HMS *Trenchant* reported hearing an unusual noise, which was later assessed to probably have been noises associated with *Antares* sinking. This noise, allied with the disappearance of *Antares'* sonar contact, should have provided sufficient evidence on board for the command team of HMS *Trenchant* to appreciate what had happened. However, there was no proper analysis of these two pieces of information and, after surfacing to clear wires from its casing, the submarine dived and continued its training exercise.

Recommendations were made in the MAIB report that included: extending the existing notification scheme, submarines not on exercise should proceed on the surface, extending the minimum passing distance for a deep submarine to 4000yds and reviewing the guidance in place at the time. The outcome of the Royal Navy's review, undertaken as a result of these recommendations, was the creation of the FV Code as described in this report.

²⁵ MAIB Report dated 15 April 1992.

1.14.2 Collision between USS *Greeneville* and *Ehime Maru*

On 9 February 2001, the United States Navy submarine, USS *Greeneville*, collided with the Japanese fishing/training vessel *Ehime Maru*. USS *Greeneville* was hosting distinguished visitors for a day at sea and was demonstrating an emergency surfacing procedure when the collision occurred. The American National Transportation Safety Board (NTSB) report²⁶ concluded that the submarine's command team failed to perform adequate contact analysis, safety procedures were rushed and the sonar teams were overloaded. These factors led to a loss of situational awareness by the submarine's command team, leading to the collision. It is evident from the NTSB report that unrestricted access to witnesses was granted to investigators, facilitating an extensive and thorough investigation of the accident.

²⁶ NTSB Case Report DCA-01-MM-022.

SECTION 2 – ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the causes and circumstances of the accident in order to make safety recommendations to prevent recurrence.

2.2 SUMMARY

Despite the Royal Navy's initial insistence that no UK or NATO submarine was within 50nm of the accident, the MAIB's initial draft investigation report concluded that *Karen* had been dragged backwards by a submarine. When the MoD later confirmed that a UK submarine had collided with *Karen*'s fishing gear, the MAIB's investigation was reopened.

The investigation focused on the circumstances that led to the collision and the actions taken afterwards. The scope of the investigation and the ability to determine all the factors that influenced the accident, particularly the decision-making on board the submarine and at the Operating Authority, was significantly constrained by the lack of access to the submarine and its crew.

Based largely on the information provided in the summary report of the Royal Navy's Directed Review, the cause of the collision and the contributing factors that led to it will be analysed in this section. The emergency response on board *Karen* and the submarine, and the actions taken by the Royal Navy after the accident, will also be discussed.

2.3 THE COLLISION

2.3.1 Misidentification of *Karen*

The submarine collided with *Karen*'s fishing gear because its sonar contact was assessed to be that of a small merchant vessel, and earlier the Commanding Officer had suspended the requirement for close quarters procedures, the effect of which was to normalise close passes with merchant vessel contacts. At the time of the collision there were 36 other vessels operating within a 10-mile radius of the submarine; most were fishing vessels and at least two²⁷ of those, including *Karen*, were within 4000yds (**Figure 19**).

Karen was misidentified, along with the majority of other fishing vessels in the area, because the submarine's sonar operators did not detect or report hearing trawl noise. Given the number of vessels operating in the area, it is almost certain that the noise levels being generated would have been extremely high, with noise from one vessel masking the noise from another. Such a situation would make it impossible for the sonar operators to methodically identify and analyse each contact, in particular to identify discrete acoustic classification clues such as trawl noise.

The misidentification of *Karen* as a merchant vessel also led directly to the TMA process over-estimating the fishing vessel's speed and, therefore, its range (**Figure 15**). The close pass with *Karen* shortly after the submarine had altered course to

²⁷ On **Figure 19**, three fishing vessels are shown within 4000 yards (2nm) of the accident location. The two positions (shown in blue) derived from AIS, including *Karen*, are timed at 1605. The position of the third fishing vessel (shown in red) also inside 4000yds of the accident location, was derived from VMS and timed at 1622 (see Section 1.11.1).

port should have alerted the command team to the fact that they had made an error with their TMA. This could have prompted them to re-evaluate the situation leading to an appreciation of a slower, closer contact. Had this happened, it is possible that the command team would have realised that they had just passed very close to a fishing vessel. However, the misidentification of *Karen* meant that the command team's perception would have been that no risk of collision could exist between a submarine at safe depth and a merchant vessel. Thus, it is assessed that, once the threat posed by *Karen* had been discounted, the command team's attention rapidly diverted to analysis of other sonar contacts.

The difficulties experienced by the submarine were the product of its speed of advance while operating deep in a densely populated fishing ground. The sonar and command teams did not have sufficient time to evaluate and classify contacts or take the necessary collision avoidance actions. The Royal Navy's Directed Review concluded that the submarine was operating near to the limit of its capability. Given that all the submarine's systems were reported to be functioning properly, it was clearly apparent that the submarine's limit of capability had, in reality, been exceeded, with its sonar and command teams becoming cognitively overloaded, leading to degraded situational awareness and poor decision-making.

2.3.2 The mechanics of the collision

There was insufficient evidence to determine with certainty the exact nature of the collision. However, it is highly likely that the initial contact by the submarine was with *Karen*'s trawl warps.

Once the submarine's forceful pull was applied to the warps, the gear will have lifted off the seabed and, in order to apply a minimum 13t force, the trawl doors, or possibly the net, must have become snagged on the submarine. Additionally, the damage to the transverse beam in the fish hold, which the winch bedplate was bolted onto, can almost certainly be attributed to the excessive, albeit brief, load on the port warp. It has not been possible to determine how the cod end bag rope was cleanly cut, but there is no evidence to suggest that this was done by the submarine or its crew.

On board *Karen*, the starboard warp was able to run freely away because it had been attached to its winch drum using rope. However, the port warp became fouled on its winch probably because the drum's rapid rotation caused the warp to ride over itself and lock²⁸. The port warp coming taut in this way, combined with the trawl doors or net being snagged on the submarine, would account for *Karen*'s uncontrolled sternway (**Table 1**). It is, therefore, extremely fortunate that the port warp parted when it did as *Karen* had already been partially submerged by the downward pull of the submarine and would inevitably have capsized and then been dragged further underwater.

Although the Royal Navy reported that the submarine was undamaged in the accident, the snagging was a very significant hazard that could potentially have disabled the submarine had its rudder, hydroplane control surfaces or propulsion system become fouled. Given that the lost fishing gear was recovered in the same fishing grounds, then it probably dropped clear of the submarine when the port warp parted; nevertheless, the potential hazard to the submarine was very significant.

²⁸ Often referred to as a 'riding turn'.

submarine's involvement to have been reported. Such a report would have clarified any assumptions being made ashore and allowed the accident investigation to commence immediately. A message from the submarine could also have confirmed that the submarine and its crew were safe after being in collision.

2.5 PLANNING AND CONDUCT OF THE PASSAGE

2.5.1 Pre-deployment planning

The Admiralty Manual of Navigation provided advice on the requirement to make an appraisal of potential navigational hazards when planning a passage. A high concentration of fishing vessels in coastal waters presents a very significant risk to the safe operation of a dived submarine and fishing vessels, which should have been identified in this process.

This risk would have been identified if the submarine's crew had examined historical fishing patterns, including AIS information, as part of the passage planning process. However, the Royal Navy's Directed Review identified that neither the command team nor the Operating Authority made sufficient reference to AIS data during the appraisal process.

Having not identified the risk of encountering high concentrations of fishing vessels, the plan, as stated at the pre-deployment brief, was to transit the Irish Sea dived at a relatively high speed through anticipated light traffic levels. This decision was not challenged by the Operating Authority but should have been as identified in the Directed Review.

If the command team's preference was to proceed deep and fast, and not be forced into situations where compliance with BR0095 would mean slowing down or operating at periscope depth, then a plan should have been made to avoid busy fishing grounds. An alternative route should have been chosen or the Irish Sea passage could have been conducted at night when fewer fishing vessels were likely to have been encountered. Alternatively, the passage should have been planned at a speed consistent with operating continuously at periscope depth.

In the days prior to sailing and even after the departure from HMNB Clyde, the submarine and the Operating Authority could have taken detailed information about fishing vessel activity into account. AIS information for the 2 days preceding the accident showed similar levels of traffic density in the Irish Sea. Had this information been considered, it would have been evident that a high concentration of fishing vessels was likely to be encountered on the planned route. Options remained continuously available to adjust the plan; however, it is evident that at no stage prior to the Irish Sea passage was the risk to the submarine and the fishing vessels properly identified by the submarine's command team or the Operating Authority.

2.5.2 Conduct of the passage in the presence of fishing vessels

When the submarine commenced its dived passage south towards the Irish Sea, it started to encounter the denser concentration of surface vessels. Had knowledge of fishing patterns and AIS information been applied to this situation, it would have been readily apparent that the majority of these contacts were vessels engaged in fishing. Therefore, it is not credible for the command team to have assessed, as was identified by the Directed Review, that, where trawl noise was absent, almost every close contact was classified as a merchant vessel.

The Directed Review identified that it was narrow and over-simplistic to use the absence of trawl noise as evidence towards an assessment that a sonar contact was not a vessel engaged in fishing. This illustrates a confirmation bias towards applying a merchant ship classification to sonar contacts; in other words, the command team wanted the surface contacts to be merchant vessels because they would present no risk of collision.

Figure 20 shows the positions of all the 61 fishing vessels²⁹ in the area, based on AIS and VMS data, at or near to the time of the accident³⁰; the circles shown around each contact are at 4000yds. It is apparent from this analysis that the area was impassable to a deep submarine operating in compliance with BR0095. It is also apparent that the submarine must have passed within 4000yds of several other fishing vessels both prior to and after the collision.

More appropriate action when the surface contact density increased would have been to assume that they were all vessels engaged in fishing unless absolutely proven otherwise. Had this happened and guidance on fishing vessel avoidance been applied, the submarine would not have pressed ahead; instead, its speed would have been reduced and it would have been returned to periscope depth. Once back at periscope depth, it would have been readily apparent - from visual and AIS information - that continuing the passage below periscope depth would be unsafe.

2.6 FISHING OPERATIONS IN SUBMARINE EXERCISE AREAS

The accident occurred within a designated submarine exercise area and, although the submarine was in transit, a SUBFACTS broadcast had warned of the presence of submarines on the day of the accident. The SUBFACTS system is intended to alert mariners to the potential presence of a submarine; it does not prohibit or restrict fishing. However, when a SUBFACTS warning is in place, fishing skippers should be guided by the advice in MGN12(F) (**Annex B**). This advice is intended to ensure the best prospect of detection by a submarine.

At the time of the accident, AIS transmissions were being detected ashore from only 40 of the 61 fishing vessels operating in the area. It is possible that the 21 vessels only identified by VMS were operating an AIS, but their transmissions were not detected by shore-based VHF aeriels; alternatively, the vessels were not fitted with an AIS system³¹. Nevertheless, this investigation has shown that AIS information is critical for submarines' situational awareness and, where fitted, AIS should be continuously transmitting to ensure this method of detection can be effectively utilised by submarines. MGN 12(F) was published before the development of AIS and would benefit from being updated to include this advice.

Additionally, the submarine was heading south out of the promulgated SUBFACTS area where no warning or notification system would have been in place. However, the FV Code applies to all areas where UK fishing vessels operate, thus it would be appropriate for fishing skippers to follow the guidance in MGN 12(F) at all times and not just when operating in designated exercise areas.

²⁹ AIS contacts identified as merchant vessels have been removed from this analysis.

³⁰ On Figure 20, fishing vessel AIS positions (shown in blue) are at the time of the accident. Given that the VMS system only transmits a fishing vessel's position approximately every 2 hours, then the fishing vessel positions derived from VMS (shown in red) are at the time nearest to 1605 that a VMS transmission was made.

³¹ A fishing vessel over 12m and below 15m in length would require a VMS but not an AIS system to be fitted (Section 1.11.1).

2.7 INVESTIGATION OF THE ACCIDENT

Learning lessons from accidents in a timely manner is key to both the prevention of recurrence and a strong safety culture. In this case, the MoD's policy of not commenting on submarine operations resulted in significant delays to the independent investigation of an accident that endangered the lives of four fishermen. Evidence provided by the Royal Navy was also insufficient to identify all the causal factors.

In response to the coastguard's initial enquiry, and in compliance with MoD policy, the duty submarine controller stated that no comment could be made regarding submarine operations. However, when the Royal Navy stated on 20 April 2015 that there was no UK or NATO submarine within 50nm of the accident location, the Operating Authority would have been aware that it was possible that a submarine might have been involved. Moreover, on 5 June 2015, when compelling evidence of submarine involvement was shared with the Royal Navy, it had a duty to establish, beyond doubt, whether a submarine was involved or not, rather than just repeating the denial.

When the presence of a UK submarine was admitted on 7 September 2015, nearly 5 months had passed since the accident, during which time perishable evidence, including witness recollections, would almost certainly have become contaminated or lost. Even after this admission, access was still denied to witnesses and other primary evidence sources required to progress the independent investigation. Instead, the MAIB had to await the outcome of the Royal Navy's Directed Review (Section 1.13) that was not made available until nearly 10 months after the accident. Although the declassified summary of this Review received by the MAIB described events leading up to the collision, it did not fully explain why it happened. Furthermore, it did not deal with why the lessons identified following the snagging of *Antares* were lost.

During this investigation, the Royal Navy has been slow to react and has actively controlled the release of evidence; this approach has impeded the MAIB's investigation. Additionally, from the information provided, the Royal Navy's own investigation did not appear to deal with the systemic issues underlying this collision and has done little to inspire confidence in its ability to react quickly and effectively to investigate and then take the actions necessary to prevent recurrence. Therefore, the Royal Navy should demonstrate that it has conducted a thorough investigation into this accident. This would ensure that the MPs' questions asked in the Westminster Hall debate were fully answered, provide assurance that lessons have been effectively managed and help rebuild trust with the fishing industry.

SECTION 3 – CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The collision happened because the submarine's command team believed *Karen* to be a merchant ship, so they did not perceive any risk of collision or need for avoiding action. [2.3.1]
2. The submarine's command team believed that *Karen* was a merchant ship primarily because no trawl noise had been heard on the same bearing. [2.3.1]
3. It is highly likely that the command team and sonar operators were cognitively overloaded due to: the density of shipping traffic; the associated noisy acoustic environment; and the unnecessarily high speed of advance. [2.3.1]
4. The circumstances of this collision were allowed to develop because the command team did not follow the standard procedures for fishing vessel avoidance set out in BR0095. Had these procedures been followed, the accident would have been avoided because the submarine would have been slowed down and returned to periscope depth when numerous vessels engaged in fishing were first encountered. [2.5.2]
5. The fishing vessel activity levels were predictable and the Irish Sea was impassable to a deep submarine operating in compliance with BR0095. This hazard should have been identified as a significant risk to the safety of both the submarine and fishing vessels during the development of the submarine's passage plan. [2.5.1]
6. If the submarine command team's priority was to achieve an undetected, deep, fast passage, then a high concentration of fishing vessels should have been avoided. [2.5.1]

3.2 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. On board the submarine, evidence of the collision was either not observed or misinterpreted. This meant that the submarine's command team was unaware of the collision until about 3 hours later. As a result, the submarine did not render immediate assistance; action that would have been necessary to preserve life had *Karen* foundered. [2.4.2]
2. Had the submarine reported its involvement in the accident, the Operating Authority would have been reassured that the submarine was safe and the accident investigation would not have been delayed. [2.4.2]
3. Delays in establishing and admitting that a submarine was responsible, as well as the Royal Navy's restrictions on the release of evidence, impeded the independent safety investigation. Additionally, the evidence submitted was insufficient to determine all the causal factors of the accident. [2.7]

4. Although there was nothing that the crew of *Karen* could have done to avoid the accident, lessons can still be learnt; in particular:
 - a Transmitting on AIS provides important information for submarines and their operating authorities ashore regarding fishing vessel activity in declared submarine exercise areas. [2.4.1]
 - b As is apparent from other accidents and incidents, raising the alarm through the use of DSC is not a natural reaction. [2.4.1]